

Лабораторная работа №2

Управление пользователями и группами

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2026-03-03

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Раздел 1

Информация

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Раздел 2

Вводная часть

Цель работы

Получить представление о работе с учётными записями пользователей и группами пользователей в операционной системе типа Linux.

- 1 Прочитайте справочное описание man по командам ls, whoami, id, groups, su, sudo, passwd, vi, visudo, useradd, usermod, userdel, groupadd, groupdel.
- 2 Выполните действия по переключению между учётными записями пользователей, по управлению учётными записями пользователей (раздел 2.4.1).
- 3 Выполните действия по созданию пользователей и управлению их учётными записями (раздел 2.4.2).
- 4 Выполните действия по работе с группами пользователей (раздел 2.4.3).

Раздел 3

Выполнение лабораторной работы

Выполнение лабораторной работы

Открываем и читаем справочники по командам ls, whoami, id, groups, su, sudo, passwd, vi, visudo, useradd, usermod, userdel, groupadd, groupdel (рис.1 - рис.14)

```
tsypinda@tsypinda:~$ man ls
tsypinda@tsypinda:~$ man whoami
tsypinda@tsypinda:~$ man id
tsypinda@tsypinda:~$ man groups
tsypinda@tsypinda:~$ man su
tsypinda@tsypinda:~$ man sudo
tsypinda@tsypinda:~$ man passwd
tsypinda@tsypinda:~$ man vi
tsypinda@tsypinda:~$ man visudo
tsypinda@tsypinda:~$ man useradd
tsypinda@tsypinda:~$ man userdel
```

```
tsypinda@tsypinda:~ -- man ls

LS(1)                                     User Commands                                     LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

    Mandatory arguments to long options are mandatory for short options too.

-a, --all
    do not ignore entries starting with .

-A, --almost-all
    do not list implied . and ..

--author
    with -l, print the author of each file

-b, --escape
    print C-style escapes for nongraphic characters

--block-size=SIZE
    with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

-B, --ignore-backups
    do not list implied entries ending with ~

Manual page ls(1) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man whoami
WHOAMI(1) User Commands WHOAMI(1)

NAME
    whoami - print effective user name

SYNOPSIS
    whoami [OPTION]...

DESCRIPTION
    Print the user name associated with the current effective user ID. Same as id -un.

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by Richard Mlynarik.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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    <https://gnu.org/licenses/gpl.html>.
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    permitted by law.

SEE ALSO
    Full documentation <https://www.gnu.org/software/coreutils/whoami>
    or available locally via: info '(coreutils) whoami invocation'
    Manual page whoami(1) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man id

ID(1)                                User Commands                                ID(1)

NAME
    id - print real and effective user and group IDs

SYNOPSIS
    id [OPTION]... [USER]...

DESCRIPTION
    Print user and group information for each specified USER, or (when USER omitted) for the current process.

    -a      ignore, for compatibility with other versions

    -Z, --context
            print only the security context of the process

    -g, --group
            print only the effective group ID

    -G, --groups
            print all group IDs

    -n, --name
            print a name instead of a number, for -ugG

    -r, --real
            print the real ID instead of the effective ID, with -ugG

    -u, --user
            print only the effective user ID

Manual page id(1) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man groups

GROUPS(1)                                User Commands                                GROUPS(1)

NAME
    groups - print the groups a user is in

SYNOPSIS
    groups [OPTION]... [USERNAME]...

DESCRIPTION
    Print group memberships for each USERNAME or, if no USERNAME is specified, for the current process
    (which may differ if the groups database has changed).

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie and James Youngman.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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    permitted by law.

SEE ALSO
    getent(1)
    Manual page groups(1) line 1 (press h for help or q to quit)
```



SU(1)

User Commands

SU(1)

NAME

su - run a command with substitute user and group ID

SYNOPSIS

su [options] [-] [user [argument...]]

DESCRIPTION

su allows commands to be run with a substitute user and group ID.

When called with no user specified, **su** defaults to running an interactive shell as root. When user is specified, additional arguments can be supplied, in which case they are passed to the shell.

For backward compatibility, **su** defaults to not change the current directory and to only set the environment variables **HOME** and **SHELL** (plus **USER** and **LOGNAME** if the target user is not root). It is recommended to always use the **--login** option (instead of its shortcut **-**) to avoid side effects caused by mixing environments.

This version of **su** uses PAM for authentication, account and session management. Some configuration options found in other **su** implementations, such as support for a wheel group, have to be configured via PAM.

su is mostly designed for unprivileged users, the recommended solution for privileged users (e.g., scripts executed by root) is to use non-set-user-ID command **runuser(1)** that does not require authentication and provides separate PAM configuration. If the PAM session is not required at all then the recommended solution is to use command **setpriv(1)**.

Note that **su** in all cases uses PAM (**pam_getenvlist(3)**) to do the final environment modification. Command-line options such as **--login** and **--preserve-environment** affect the environment before it is modified by PAM.

Manual page su(1) line 1 (press h for help or q to quit)



```
tsypinda@tsypinda:~ -- man sudo
SUDO(8)                                     System Manager's Manual                                     SUDO(8)

NAME
    sudo, sudoedit - execute a command as another user

SYNOPSIS
    sudo -h | -K | -k | -V
    sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
    sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user] [-u user] [command [arg ...]]
    sudo [-ABbEHnPS] [-C num] [-D directory] [-g group] [-h host] [-p prompt] [-R directory] [-r role]
        [-t type] [-T timeout] [-u user] [VAR=value] [-i | -s] [command [arg ...]]
    sudoedit [-ABkNnS] [-C num] [-D directory] [-g group] [-h host] [-p prompt] [-R directory] [-r role]
        [-t type] [-T timeout] [-u user] file ...

DESCRIPTION
    sudo allows a permitted user to execute a command as the superuser or another user, as specified by
    the security policy. The invoking user's real (not effective) user-ID is used to determine the user
    name with which to query the security policy.

    sudo supports a plugin architecture for security policies, auditing, and input/output logging. Third
    parties can develop and distribute their own plugins to work seamlessly with the sudo front-end. The
    default security policy is sudoers, which is configured via the file /etc/sudoers, or via LDAP. See
    the "Plugins" section for more information.

    The security policy determines what privileges, if any, a user has to run sudo. The policy may re-
    quire that users authenticate themselves with a password or another authentication mechanism. If au-
    thentication is required, sudo will exit if the user's password is not entered within a configurable
    time limit. This limit is policy-specific; the default password prompt timeout for the sudoers secu-
    rity policy is 5 minutes.

    Security policies may support credential caching to allow the user to run sudo again for a period of
    time without requiring authentication. By default, the sudoers policy caches credentials on a per-
    Manual page sudo(8) line 1 (press h for help or q to quit)
```

```
+ tsypinda@tsypinda:~ -- man passwd
PASSWD(1) User Commands PASSWD(1)

NAME
    passwd - change user password

SYNOPSIS
    passwd [options] [LOGIN]

DESCRIPTION
    The passwd command changes passwords for user accounts. A normal user may only change the password for their own account, while the superuser may change the password for any account. passwd also changes the account or associated password validity period.

    Password Changes
    The user is first prompted for their old password, if one is present. This password is then encrypted and compared against the stored password. The user has only one chance to enter the correct password. The superuser is permitted to bypass this step so that forgotten passwords may be changed.

    After the password has been entered, password aging information is checked to see if the user is permitted to change the password at this time. If not, passwd refuses to change the password and exits.

    The user is then prompted twice for a replacement password. The second entry is compared against the first and both are required to match in order for the password to be changed.

    Then, the password is tested for complexity. passwd will reject any password which is not suitably complex. Care must be taken not to include the system default erase or kill characters.

    Hints for user passwords
    The security of a password depends upon the strength of the encryption algorithm and the size of the key space. The legacy UNIX System encryption method is based on the NBS DES algorithm. More recent

Manual page passwd(1) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man vi
VIM(1)                                General Commands Manual                                VIM(1)

NAME
    vim - Vi IMproved, a programmer's text editor

SYNOPSIS
    vim [options] [file ..]
    vim [options] -
    vim [options] -t tag
    vim [options] -q [errorfile]

    ex gex
    view
    gvim gview vimx evim eview
    rvim rview rgvim rgview

DESCRIPTION
    Vim is a text editor that is upwards compatible to Vi. It can be used to edit all kinds of plain text. It is especially useful for editing programs.

    There are a lot of enhancements above Vi: multi level undo, multi windows and buffers, syntax highlighting, command line editing, filename completion, on-line help, visual selection, etc.. See ":help vi_diff.txt" for a summary of the differences between Vim and Vi.

    While running Vim a lot of help can be obtained from the on-line help system, with the ":help" command. See the ON-LINE HELP section below.

    Most often Vim is started to edit a single file with the command

        vim file

    More generally Vim is started with:
    Manual page vi(1) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man visudo
VISUDO(8)                                     System Manager's Manual                                     VISUDO(8)

NAME
    visudo - edit the sudoers file

SYNOPSIS
    visudo [-chIOPqsV] [[-f] sudoers]

DESCRIPTION
    visudo edits the sudoers file in a safe fashion, analogous to vi(8). visudo locks the sudoers file against multiple simultaneous edits, performs basic validity checks, and checks for syntax errors before installing the edited file. If the sudoers file is currently being edited you will receive a message to try again later.

    If the sudoers file does not exist, it will be created unless the editor exits without writing to the file.

    visudo parses the sudoers file after editing and will not save the changes if there is a syntax error. Upon finding an error, visudo will print a message stating the line number(s) where the error occurred and the user will receive the "What now?" prompt. At this point the user may enter 'e' to re-edit the sudoers file, 'x' to exit without saving the changes, or 'Q' to quit and save changes. The 'Q' option should be used with extreme caution because if visudo believes there to be a syntax error, so will sudo. If 'e' is typed to edit the sudoers file after a syntax error has been detected, the cursor will be placed on the line where the error occurred (if the editor supports this feature).

    There are two sudoers settings that determine which editor visudo will run.

    editor      A colon (':') separated list of editors allowed to be used with visudo. visudo will choose the editor that matches the user's SUDO_EDITOR, VISUAL, or EDITOR environment variable if possible, or the first editor in the list that exists and is executable. sudo does not preserve the SUDO_EDITOR, VISUAL, or EDITOR environment variables unless they are present in the env_keep list or the env_reset option is disabled in the sudoers file. The
```

Manual page visudo(8) line 1 (press h for help or q to quit)

```
tsypinda@tsypinda:~ -- man useradd
USERADD(8)                                System Management Commands                                USERADD(8)

NAME
    useradd - create a new user or update default new user information

SYNOPSIS
    useradd [options] LOGIN

    useradd -D

    useradd -D [options]

DESCRIPTION
    When invoked without the -D option, the useradd command creates a new user account using the values specified on the command line plus the default values from the system. Depending on command line options, the useradd command will update system files and may also create the new user's home directory and copy initial files.

    By default, a group will also be created for the new user (see -g, -N, -U, and USERGROUPS_ENAB).

OPTIONS
    The options which apply to the useradd command are:

    --badname
        Allow names that do not conform to standards.

    -b, --base-dir BASE DIR
        The default base directory for the system if -d HOME DIR is not specified. BASE DIR is concatenated with the account name to define the home directory.

        If this option is not specified, useradd will use the base directory specified by the HOME

Manual page useradd(8) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man userdel

USERDEL(8)                                System Management Commands                                USERDEL(8)

NAME
    userdel - delete a user account and related files

SYNOPSIS
    userdel [options] LOGIN

DESCRIPTION
    The userdel command modifies the system account files, deleting all entries that refer to the user name LOGIN. The named user must exist.

OPTIONS
    The options which apply to the userdel command are:

    -f, --force
        This option forces the removal of the user account, even if the user is still logged in. It also forces userdel to remove the user's home directory and mail spool, even if another user uses the same home directory or if the mail spool is not owned by the specified user. If USERGROUPS_ENAB is defined to yes in /etc/login.defs and if a group exists with the same name as the deleted user, then this group will be removed, even if it is still the primary group of another user.

        Note: This option is dangerous and may leave your system in an inconsistent state.

    -h, --help
        Display help message and exit.

    -r, --remove
        Files in the user's home directory will be removed along with the home directory itself and the user's mail spool. Files located in other file systems will have to be searched for and deleted manually.

Manual page userdel(8) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man groupadd

GROUPADD(8)                                System Management Commands                                GROUPADD(8)

NAME
    groupadd - create a new group

SYNOPSIS
    groupadd [OPTIONS] NEWGROUP

DESCRIPTION
    The groupadd command creates a new group account using the values specified on the command line plus the default values from the system. The new group will be entered into the system files as needed.

    Groupnames may contain only lower and upper case letters, digits, underscores, or dashes. They can end with a dollar sign. Dashes are not allowed at the beginning of the groupname. Fully numeric groupnames and groupnames . or .. are also disallowed.

    Groupnames may only be up to 32 characters long.

OPTIONS
    The options which apply to the groupadd command are:

    -f, --force
        This option causes the command to simply exit with success status if the specified group already exists. When used with -g, and the specified GID already exists, another (unique) GID is chosen (i.e. -g is turned off).

    -g, --gid GID
        The numerical value of the group's ID. GID must be unique, unless the -o option is used. The value must be non-negative. The default is to use the smallest ID value greater than or equal to GID_MIN and greater than every other group.

Manual page groupadd(8) line 1 (press h for help or q to quit)
```

```
tsypinda@tsypinda:~ -- man groupdel

GROUPDEL(8)                                System Management Commands                                GROUPDEL(8)

NAME
    groupdel - delete a group

SYNOPSIS
    groupdel [options] GROUP

DESCRIPTION
    The groupdel command modifies the system account files, deleting all entries that refer to GROUP. The
    named group must exist.

OPTIONS
    The options which apply to the groupdel command are:

    -f, --force
        This option forces the removal of the group, even if there's some user having the group as the
        primary one.

    -h, --help
        Display help message and exit.

    -R, --root CHROOT_DIR
        Apply changes in the CHROOT_DIR directory and use the configuration files from the CHROOT_DIR
        directory. Only absolute paths are supported.

    -P, --prefix PREFIX_DIR
        Apply changes in the PREFIX_DIR directory and use the configuration files from the PREFIX_DIR
        directory. This option does not chroot and is intended for preparing a cross-compilation target.
        Some limitations: NIS and LDAP users/groups are not verified. PAM authentication is using the host
        files. No SELINUX support.

Manual page groupdel(8) line 1 (press h for help or q to quit)
```

Просматриваем более подробную информацию с помощью id (рис.16)

```
tsypinda@tsypinda:~$ id
uid=1000(tsypinda) gid=1000(tsypinda) groups=1000(tsypinda),10(wheel) context=un
confined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
tsypinda@tsypinda:~$
```

Рисунок 16: id

Переключаем учетную запись в root и повторно смотрим информацию (рис.17). В строчке каждое значение значит следующее: uid=0 - id user'а, в данном случае 0 т.к. root gid=0 - id группы, аналогично 0 т.к. root groups - группы root

```
tsypinda@tsypinda:~$ su
Password:
root@tsypinda:/home/tsypinda# id
uid=0(root) gid=0(root) groups=0(root) context=unconfined_u:unconfined_r:unconfi
ned_t:s0-s0:c0.c1023
root@tsypinda:/home/tsypinda#
```

Рисунок 17: su, id

Возвращаемся к изначальной учетной записи (рис.18)

```
root@tsypinda:/home/tsypinda# exit  
exit  
tsypinda@tsypinda:~$
```

Рисунок 18: exit

Просматриваем в безопасном режиме файл /etc/sudoers (рис.19 - рис.20)

```
tsypinda@tsypinda:~$ sudo -i visudo
```

Рисунок 19: sudo -i visudo

```
## Sudoers allows particular users to run various commands as
## the root user, without needing the root password.
##
## Examples are provided at the bottom of the file for collections
## of related commands, which can then be delegated out to particular
## users or groups.
##
## This file must be edited with the 'visudo' command.

## Host Aliases
## Groups of machines. You may prefer to use hostnames (perhaps using
## wildcards for entire domains) or IP addresses instead.
# Host_Alias      FILESERVERS = fs1, fs2
# Host_Alias      MAILSERVERS = smtp, smtp2
```

Находим строчку %wheel ALL=(ALL) ALL (рис.21)

```
## Allows people in group wheel to run all commands  
%wheel  ALL=(ALL)      ALL
```

Рисунок 21: %wheel ALL=(ALL) ALL

Создаем пользователя alice с группой wheel. С помощью id убеждаемся в правильности созданного пользователя (рис.22)

```
tsypinda@tsypinda:~$ sudo -i useradd -G wheel alice
tsypinda@tsypinda:~$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),10(wheel)
tsypinda@tsypinda:~$ █
```

Рисунок 22: sudo -i useradd -G wheel alice

Задаем пароль пользователю alice (рис.23)

```
passwd: password unchanged  
tsypinda@tsypinda:~$ sudo -i passwd alice  
New password:  
BAD PASSWORD: The password is a palindrome  
Retype new password:  
passwd: password updated successfully  
tsypinda@tsypinda:~$
```

Рисунок 23: sudo -i passwd alice

Переключаемся на alice (рис.24)

```
tsypinda@tsypinda:~$ su alice  
Password:  
alice@tsypinda:/home/tsypinda$
```

Рисунок 24: su alice

Создаем пользователя bob (рис.25) и задаем ему пароль (рис. 26)

```
alice@tsypinda:/home/tsypinda$ sudo useradd bob
```

We trust you have received the usual lecture from the local System Administrator. It usually boils down to these three things:

- #1) Respect the privacy of others.
- #2) Think before you type.
- #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

```
[sudo] password for alice:  
alice@tsypinda:/home/tsypinda$
```

Рисунок 25: sudo useradd bob

```
alice@tsypinda:/home/tsypinda$ sudo passwd bob  
New password:
```

Просматриваем, в каких группах находится bob (рис. 27)

```
alice@tsypinda:/home/tsypinda$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)
alice@tsypinda:/home/tsypinda$
```

Рисунок 27: id bob

Переключаемся в рут и открываем файл конфигурации /etc/login.defs для редактирования (рис.28)

```
root@tsypinda:/home/tsypinda# su
root@tsypinda:/home/tsypinda# vim /etc/login.defs
root@tsypinda:/home/tsypinda#
```

Рисунок 28: vim /etc/login.defs

Проверяем параметр CREATE_HOME (рис.29), меняем USERGROUPS_ENAB на no (рис.30)

```
#  
CREATE_HOME yes  
#
```

Рисунок 29: CREATE_HOME

```
#  
# Enables userdel(8) to remove user groups if no members exist.  
#  
USERGROUPS_ENAB no
```

Переходим в каталог /etc/skel и создаем каталоги Pictures и Documents (рис.31)

```
root@tsypinda:/home/tsypinda# cd /etc/skel
root@tsypinda:/etc/skel# mkdir Pictures
root@tsypinda:/etc/skel# mkdir Documents
root@tsypinda:/etc/skel#
```

Рисунок 31: Переход в каталог /etc/skel, создание каталогов Pictures и Documents

Открываем для редактирования файл `.bashrc` (рис.32), добавляем строку `export EDITOR=/usr/bin/vim` (рис.33)

```
root@tsypinda:/etc/skel# sudo vim .bashrc
root@tsypinda:/etc/skel#
```

Рисунок 32: `sudo vim .bashrc`

```
export EDITOR=/usr/bin/vim
".bashrc" 26L, 549B
```

Рисунок 33: `export EDITOR=/usr/bin/vim`

Переключаемся на учетную запись alice, создаем пользователя carol. Устанавливаем пароль (рис.34)

```
root@tsypinda:/etc/skel# su alice
alice@tsypinda:/etc/skel$ sudo -i useradd carol
[sudo] password for alice:
alice@tsypinda:/etc/skel$ sudo passwd carol
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: password updated successfully
alice@tsypinda:/etc/skel$ █
```

Рисунок 34: Создание carol

Переключаемся на carol. Проверяем первоначальную группу (100(users)), убеждаемся в наличии каталогов documentns и Pictures (рис.35)

```
alice@tsypinda:/etc/skel$ su carol
Password:
carol@tsypinda:/etc/skel$ id
uid=1003(carol) gid=100(users) groups=100(users) context=unconfined_u:unconfined
_r:unconfined_t:s0-s0:c0.c1023
carol@tsypinda:/etc/skel$ cd
carol@tsypinda:~$ ls -Al
total 12
-rw-r--r--. 1 carol users 18 Oct 29 2024 .bash_logout
-rw-r--r--. 1 carol users 144 Oct 29 2024 .bash_profile
-rw-r--r--. 1 carol users 549 Sep 11 13:36 .bashrc
drwxr-xr-x. 2 carol users  6 Sep 11 13:35 Documents
drwxr-xr-x. 4 carol users 39 Sep  9 18:35 .mozilla
drwxr-xr-x. 2 carol users  6 Sep 11 13:35 Pictures
carol@tsypinda:~$
```

Рисунок 35: Проверка carol

Переключаемся на alice и вызываем строку записи о пароле carol (рис.36). В этой строке разделение показывается с помощью :. Каждая часть означает следующее: 1. Имя пользователя

2. Пароль в зашифрованном виде

3. Количество дней с 1 января 1970 года до момента создания пользователя

4. Количество дней, после которых можно будет сменить пароль

5. Количество дней, через которые пароль станет недействительным

6. Количество дней, за сколько система предупредит пользователя о том, что его пароль вскоре станет недействительным

7. Количество дней, по прошествии которых пользователь станет недействительным 8.

Сколько дней прошло после того, как пользователь стал недействительным

9. Дополнительная секция

```
carol@tsypinda:~$ su alice
Password:
alice@tsypinda:/home/carol$ sudo cat /etc/shadow | grep carol
carol:$y$j9T$MaMECmI7WTU8X01RCTJVV.$/.fyBTzLS/.qm.Dv70/tXyy3vgx18VuYQw6ULAzJEKA:
20707:0:99999:7:::
alice@tsypinda:/home/carol$
```

Изменяем свойства пароля carol (рис.37)

```
alice@tsypinda:/home/carol$ sudo passwd -n 30 -w 3 -x 90 carol
[sudo] password for alice:
passwd: password changed.
alice@tsypinda:/home/carol$ sudo cat /etc/shadow | grep carol
carol:$y$j9T$MaMECmI7WTU8X01RCTJVV.$/.fyBTzLS/.qm.Dv70/tXyy3vgx18VuYQw6ULAzJEKA:
20707:30:90:3:::
alice@tsypinda:/home/carol$
```

Рисунок 37: sudo passwd -n 30 -w 3 -x 90 carol

Убеждаемся, что идентификатор alice существует во всех трех файлах (рис.38)

```
alice@tsypinda:/home/carol$ su
Password:
root@tsypinda:/home/carol# grep alice /etc/passwd /etc/shadow /etc/group
/etc/passwd:alice:x:1001:1001::/home/alice:/bin/bash
/etc/shadow:alice:$y$j9T$gEC7HngsKDBR8gk19hn0P1$ORRBei.VE/CciaHvN.zzpWF0dr0A9435
2q9f/jPBV.1:20707:0:99999:7:::
/etc/group:wheel:x:10:tsypinda,alice
/etc/group:alice:x:1001:
root@tsypinda:/home/carol#
```

Рисунок 38: grep alice /etc/passwd /etc/shadow /etc/group

Убеждаемся, что идентификатор carol существует не во всех трех файлах (рис. 39)

```
root@tsypinda:/home/carol# grep carol /etc/passwd /etc/shadow /etc/group
/etc/passwd:carol:x:1003:100::/home/carol:/bin/bash
/etc/shadow:carol:$y$j9T$MaMECmI7WTU8X01RCTJVV.$/.fyBTzLS/.qm.Dv70/tXyy3vgx18VuY
Qw6ULAzJEKA:20707:30:90:3:::
root@tsypinda:/home/carol#
```

Рисунок 39: sudo grep carol /etc/passwd /etc/shadow /etc/group

Переходим в alice и создаем группы main third (рис.40)

```
root@tsypinda:/home/carol# su alice
alice@tsypinda:/home/carol$ sudo groupadd main
[sudo] password for alice:
alice@tsypinda:/home/carol$ sudo groupadd third
alice@tsypinda:/home/carol$ █
```

Рисунок 40: sudo groupadd main, sudo groupadd third

Создаем пользователей dave, dan и david, а также задаем им пароли (рис. 41)

```
alice@tsypinda:/home/carol$ sudo useradd dave
alice@tsypinda:/home/carol$ sudo passwd dave
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: password updated successfully
alice@tsypinda:/home/carol$ sudo useradd dan
alice@tsypinda:/home/carol$ sudo useradd david
alice@tsypinda:/home/carol$ sudo passwd dan
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: password updated successfully
alice@tsypinda:/home/carol$ sudo passwd david
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: password updated successfully
```

Добавляем пользователей в группы (рис.42)

```
alice@tsypinda:/home/carol$ sudo usermod -aG main alice
alice@tsypinda:/home/carol$ sudo usermod -aG main bob
alice@tsypinda:/home/carol$ sudo usermod -aG third carol
alice@tsypinda:/home/carol$ sudo usermod -aG third dan
alice@tsypinda:/home/carol$ sudo usermod -aG third dave
alice@tsypinda:/home/carol$ sudo usermod -aG third david
alice@tsypinda:/home/carol$ █
```

Рисунок 42: sudo usermod -aG группа пользователь

Убеждаемся, что carol правильно добавлен в группу third (рис.43)

```
alice@tsypinda:/home/carol$ id carol  
uid=1003(carol) gid=100(users) groups=100(users),1004(third)  
alice@tsypinda:/home/carol$
```

Рисунок 43: id carol

Определяем, участниками каких групп являются другие созданные пользователи (рис.44)

```
alice@tsypinda:/home/carol$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),10(wheel),1003(main)
alice@tsypinda:/home/carol$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1003(main)
alice@tsypinda:/home/carol$ id dan
uid=1005(dan) gid=100(users) groups=100(users),1004(third)
alice@tsypinda:/home/carol$ id dave
uid=1004(dave) gid=100(users) groups=100(users),1004(third)
alice@tsypinda:/home/carol$ id david
uid=1006(david) gid=100(users) groups=100(users),1004(third)
alice@tsypinda:/home/carol$ █
```

Рисунок 44: Другие пользователи

Раздел 4

Выводы

Я получил представление о работе с учётными записями пользователей и группами пользователей в операционной системе типа Linux.

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